

.NET MCQ

1. What happens when you compile this code?

```
var query = from e in employees
             where e.Age > 30
             select e.Name, e.Age;
```

- A. It works fine and returns Name and Age together.
 - B. It only works if you write using System.Tuple;.
 - C. You can replace the comma with into to fix it.
 - D. It gives an error because select must return a single object.
 - Answer: D
-

2. You are working on a banking application that saves **Customer** objects to disk. The **Customer** class is given below. When serializing a **Customer** object, which field/property will **NOT** be saved to disk?

```
[Serializable]
public class Customer
{
    public string Name { get; set; }
    public int Age { get; set; }
    [NonSerialized] private string _password;
    public string TemporarySessionToken { get; set; }
}
```

- A. Name
 - B. _password & TemporarySessionToken
 - C. _password
 - D. TemporarySessionToken
 - Answer: C
-

3. What happens when you compile this code?

```
public class ConfigLoader {
    var _config = LoadConfig(); // Line 1
    public void Init() {
```

```
var config = _config;    // Line 2
var = "test";            // Line 3
var timeout = 30;        // Line 4
Console.WriteLine(timeout.GetType().Name);
}
private Dictionary<string, string> LoadConfig() => new();
}
```

- A. Output: Int32
 - B. Compile error at Line 1: "var cannot be used here" and Line 3: "Invalid expression"
 - C. Output: Dictionary`2
 - D. Compile error at Line 1 and runtime error at Line 4
 - Answer: B
-

4. Which of the following statements about module initializers is FALSE?

- A. Module initializers execute before any other method in the assembly, including static constructors.
 - B. A single assembly can contain multiple module initializer methods.
 - C. Module initializers can be defined as instance methods in non-static classes.
 - D. If a module initializer throws an exception, the application fails immediately
 - Answer: C
-

5. With nullable reference types enabled What does the compiler do?

```
string nonNullableString = "Hello";
var inferredString = nonNullableString;
inferredString = null;
```

- A. No warnings – var infers non-nullable string.
 - B. Warning – var infers string? (nullable).
 - C. Error – var does not allow null assignments.
 - D. No warnings – var acts like dynamic.
 - Answer: A
-

6. In a WinForms app, the auto-generated MainForm.Designer.cs declares a partial method:

- partial void InitializeCustomComponents();
- A developer implements it in MainForm.cs:

```
partial void InitializeCustomComponents() {  
    // ...  
}
```

- Later, another developer adds a second implementation in MainForm.Experimental.cs. What happens?
 - A. Compile error – Partial methods allow only one implementation
 - B. Runtime error – Duplicate method execution
 - C. Silent merge – Both implementations run
 - D. Warning only – Code works with a compiler warning
 - Answer: A
-

7. Which of the following statements about extension methods is true?

- A. An extension method declared in a static class can override an instance method on the same type if it has the same signature.
 - B. Extension methods can be declared as static methods in any class, even if the class itself is not static, as long as the method is static and public.
 - C. If an extension method and an instance method have the same name and signature, the compiler will always prefer the extension method.
 - D. Extension methods must be static methods declared inside a top-level static class, and the first parameter must be prefixed with "this" to specify the type they extend.
 - Answer: D
-

8. What will be the output of the following code snippet?

```
class Program {  
    static void Main(string[] args) {  
        int[] nums = {3, 1, 2, 5, 4};  
        var ltAvg = from n in nums  
                     let x = nums.Average()  
                     where n < x  
                     select n;  
        Console.WriteLine("The average is " + nums.Average());  
    }  
}
```

- A. Run time error

- B. 3
 - C. 1, 2
 - D. Compile time error
 - Answer: B
-

9. Consider this StreamWriter constructor - Which statement about the parameters is FALSE?

◦ new StreamWriter(string path, bool append)

- A. path must be a full absolute path – relative paths always throw ArgumentException
 - B. When append is true, the writer preserves original content but does not guarantee atomic writes.
 - C. If the file at path is locked by another process, this will throw IOException regardless of append value.
 - D. Path can be a UNC network path (e.g., `\\server\file.txt`).
 - Answer: A
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10. Which of the following statements about the ArgumentException class is correct?

- A. ArgumentException is thrown when a method receives a null argument that is not allowed; therefore, it is the same as ArgumentNullException.
- B. ArgumentException is in the System.Collections namespace.
- C. ArgumentException indicates that at least one of the arguments passed to a method does not meet the parameter requirements.
- D. ArgumentException can only be thrown by the .NET runtime and cannot be thrown explicitly in your code.
- Answer: C